Muhammad Haider Zaidi

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Hiring Manager, Deep Learning & Algorithms Team
Abbott
Munich, Germany
May 14, 2026

Re: Application for Software Engineer (m/f/d) Deep Learning and Algorithms - heart.ai

I am writing to express my enthusiastic interest in the Software Engineer (m/f/d) Deep Learning and Algorithms - heart.ai position at Abbott in Munich. Abbott's commitment to breakthrough science and improving people's health deeply resonates with my passion for leveraging advanced AI and deep learning to solve complex challenges. The opportunity to contribute to innovative solutions like heart.ai, supporting cardiac interventions through cutting-edge algorithms and medical imaging, presents an exciting alignment with my career aspirations and expertise.

My professional background includes extensive experience in developing and optimizing deep learning models and algorithms, directly applicable to the challenges at heart.ai. At Folio3, I engineered advanced image segmentation algorithms, notably improving vitiligo segmentation accuracy to 94% and increasing processing speed by 5x, an experience highly relevant to 3D segmentation for medical imaging. I further optimized model performance and reduced compute costs by 6x for real-time applications using TensorRT, while managing MLOps pipelines on AWS SageMaker, addressing data drift and maintaining Kubernetes infrastructure. This provided hands-on experience with AWS-based cloud products and ensuring robust, efficient model deployment. At the German National Metrology Institute (PTB), I designed and evaluated novel deep learning architectures and physics-informed hybrid loss functions, focusing on improving model accuracy and stability for metrological applications. My work involved continuous improvement of model performance and training efficiency, aligning with your emphasis on rigorous validation and algorithmic robustness. Furthermore, my collaboration with PhD researchers for a FLOMEKO conference publication demonstrates my ability to contribute to research and work effectively within a scientific team.

I am particularly drawn to Abbott's mission of fostering a healthier life and the dynamic, collaborative environment within the heart.ai team. My strong communication skills, structured problem-solving capabilities, and high sense of ownership, honed through collaborating with diverse technical and clinical teams, make me confident in my ability to quickly integrate and contribute meaningfully. I am eager to discuss how my expertise in deep learning, algorithms, and robust software development can support Abbott's pioneering work in cardiac interventions. Thank you for your time and consideration.

Sincerely,

Muhammad Haider Zaidi